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SUPERSEDING
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April 9, 1954

FEDERAL SPECIFICATION

VISES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers bench and hand vises generally used in metalworking and woodworking shops.

1.1.1 *Federal specification coverage.*—This specification covers only the types, classes, and styles of the commodity generally purchased by the Federal Government and does not include all types, classes, and styles which are commercially available.

1.2 Classification.

1.2.1 *Types, classes, and styles.* — Vises shall be of the following types, classes, and styles, as specified (see 6.2) :

Type I.—Bench, clamp base (utility).
  Class 1.—Stationary base.
  Class 2.—Swivel base.

Type II.—Blacksmith's.

Type III.—Sheetmetal worker's.

Type IV.—Bench and pipe (swivel base, stationary jaw).

Type V.—Hand.
  Class 1.—Hinged jaw.
  Class 2.—Hollow handle.
  Class 3.—Combination bench and hand.
    Style A.—Without bench clamp.
    Style B.—With bench clamp.

Type VI.—Jeweler's.
  Class 1.—Stationary base.
  Class 2.—Swivel base.

Type VII.—Machinist's.
  Class 1. — Stationary base, stationary jaw.
  Class 2.—Swivel base, stationary jaw.
  Class 3.—Swivel base, swivel jaw.

Type IX.—Pin.
  Class 1.—Metal handle.
  Class 2.—Nonmetallic handle.

Type X.—Pipe.
  Class 1.—Chain.
  Class 2.—Hinged jaw.
  Class 3.—Open side.

Type XI.—Wire rope splicing.

Type XII.—Handsaw filing.
  Class 1.—With filing guide.
  Class 2.—Without filing guide.

Type XIII.—Band saw filing.

Type XIV.—Circular saw filing.

Type XV.—Woodworker's.
  Class 1.—Continuous screw.
  Class 2.—Quick acting.

Type XVI.—Patternmaker's.

Type XVII.—Machine table.
  Class 1.—Toolmaker's.
    Style A.—Slotted base.
    Style B.—Solid base.
  Class 2.—Drill.
    Style A.—Quick action.

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Style B. — Continuous screw, light duty.

Style C. — Continuous screw, heavy duty.

Type XVIII.—Bench and machine table.

Type XIX.—Contour gripping.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Nongovernmental:

Official Classification Committee Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of vises otherwise conforming to this specification.

3.2 Materials.

3.2.1 Materials used in the construction of vises shall be as hereinafter specified for each type and class. All castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength and other essential physical properties of the casting shall be adequate throughout for the purpose intended. When materials are used which are

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not specifically designated, they shall be entirely suitable for the purpose intended.

3.2.2 *Coatings.*—All unmachined surfaces of the component parts of vises shall be given either one coat of baked enamel or lacquer or one coat of primer and one coat of air-drying enamel or lacquer except type X vise, which may have an oil coating as an alternate. Unless otherwise specified (see 6.2), the color may be at the option of the contractor. In no case shall fillers be used.

3.3 Identification marking.—Each vise shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.4 Interchangeability of parts.—All parts having the same manufacturer's number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.5 Vise screw.—The vise screw shall be made from steel of not less than 60,000 pounds per square inch (p.s.i.) yield and 80,000 p.s.i. tensile strength (see 4.5.3).

3.5.1 *Screw threads.*—The screw threads used on vises shall be in accordance with National Bureau of Standards Handbook H28 and supplement thereto.

3.5.2 *Jaw caps.*—When specified (see 6.2), jaw caps of solid brass, copper, aluminum, aluminum bronze alloy, plastic, or smooth soft steel shall be furnished.

3.6 Type I, bench, clamp base (utility).—Type I vise shall consist of a body with back jaw, front jaw, vise screw, screw handle, and bench-clamping screw. The vise shall be provided with guide rods, a single rectangular steel guide bar or vise nut and sliding member. When guide rods are furnished, they shall be of steel and rigidly attached to the front jaw. The axis of the rods shall be parallel to the axis of the screw. When specified (see 6.2), the vise shall have pockets provided for pipe jaws to hold round work.

3.6.1 *Body.*—The body shall be of cast iron or malleable iron. The face of the back jaw shall be furnished square and perpendicular to the base within a tolerance of 1 degree in either direction (see 4.3.1). The bottom surface of the body base shall be solid in cross section. When guide rods are furnished, the body shall be threaded to receive the vise screw. The length of the threaded bearing portion of the body shall be not less than one and one-half times the diameter of the screw.

3.6.2 *Front jaw.*—The front jaw shall be of the same material as the body. When front jaws and sliding members are furnished, the sliding member shall conform to the requirements specified in 3.8.2 except the sliding member shall be finished to allow not more than 0.008-inch clearance at any point between the sliding member and its counterpart. The face of the front jaw shall remain parallel to the face of the back jaw in any position of travel.

3.6.3 *Vise screw.*—The vise screw shall have American National, square, acme, modified acme, or buttress threads and be in accordance with 3.5. One end of the screw shall have a steel head of adequate size which shall be either an integral part of the screw or welded to the screw in a permanent manner. The head shall be provided with a hole to receive the screw handle. The hole for the screw handle shall be chamfered at both ends to prevent accidental pinching of the hands. The screw shall engage the full length of the threaded bearing of the body when the jaw is at the maximum opening indicated in the applicable table.

3.6.4 *Screw handle.* — The screw handle shall be a straight, round steel bar of a diameter comparable to the size of the vise. The screw handle shall slide freely in the hole of the screwhead with means provided at each end of the handle to prevent complete disengagement from the screwhead.

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3.6.5 *Bench-clamping screw.*—The bench-clamping screw shall be of a commercial-steel screwstock with either a knob wing "T" or sliding bar handle. The clamping screw shall be capable of permitting the vise to be securely tightened to bench tops which vary in thickness from ½ to 2 inches. The upper or clamping end of the screw shall have a swivel button to compensate for any irregularities of the bench. The screw threads shall conform to 3.5.1.

3.6.6 Type I, classes 1 and 2 vise shall conform to table I for the size specified (see 6.2).

3.6.7 *Class 1, stationary base.*—The front of the body shall extend down and form a yoke section with the bottom surface of the body to provide a bench-clamping arrangement. Class 1 vise shall be similar to figure 1.

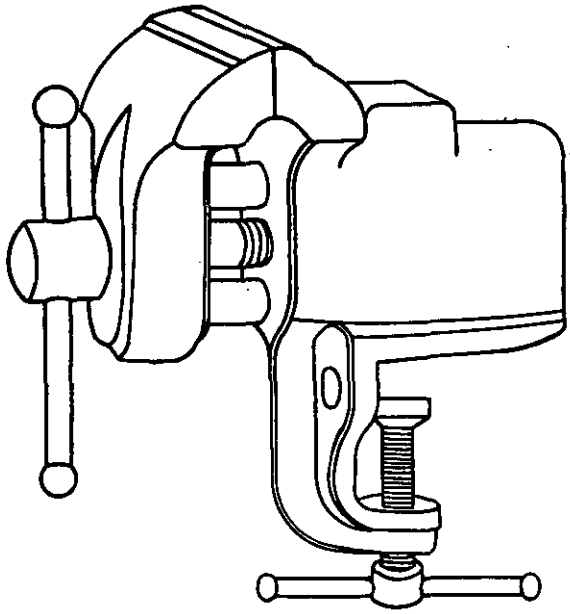


FIGURE 1.—Type I, class 1, bench, clamp base (utility), stationary base.

3.6.8 *Class 2, swivel base.*—Class 2 vise shall be similar to figure 2, and shall be designed to swivel on a swivel-base plate. The vise, when mounted on the plate, shall swivel through 360 degrees and shall have a positive locking arrangement to secure the vise in any angular position.

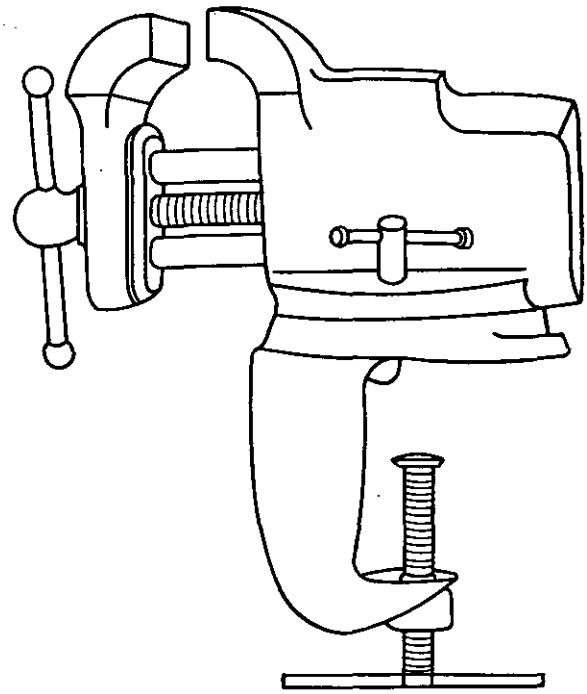


FIGURE 2.—Type I, class 2, bench, clamp base (utility), swivel base.

3.7 **Type II, blacksmith's.**—The vise shall consist of a body with back jaw and floor rod, front jaw, nut, vise screw, screw handle, spring, and pivot link.

3.7.1 *Body.*—The body shall be of forged steel or wrought iron and shall be integral with the back jaw and floor rod. The body shall be threaded to receive the vise screw or arranged to accommodate a nut. The nut, if used, shall be designed to fit in the body in such a manner as not to revolve when screw is rotated in opening and closing the jaws. The body shall be provided with a bench mounting plate or other means to securely fasten the vise to a bench in such a manner that the vise shall not become loose under ordinary service conditions.

3.7.2 *Front jaw.*—The front jaw shall be of the same material as the body. A pivot link and pivot pin or other suitable means shall be provided for attaching the front jaw to the body. The face of the front jaw shall remain parallel to the face of the back jaw in any position of travel.

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TABLE I.—*Type I, classes 1 and 2, bench, clamp base (utility)*

Jaw width (nominal) ±1/8 inch	Jaw opening (minimum)	Throat depth (minimum)	Weight (minimum)	
			Class 1	Class 2
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
1½	1½	1	2¼	3¼
2	2	1	3¾	4¾
2½	2¼	1¼	4	5
3	3	1¼	7	10

3.7.3 *Nut.*—The nut, when not an integral part of the body, shall be of malleable iron or steel, threaded internally to engage the thread of the screw, and shall be at least two and one-half times the major diameter of the screw in length. The nut shall fit properly in the body so that it is in alignment with the screw and there is no binding of the screw in the nut when the jaws of the vise are opened. Any clearance between the nut and the body should not exceed 0.010 inch.

3.7.4 *Vise screw.*—The vise screw shall conform to the requirements specified in 3.6.3 except the screw shall have square, acme, or buttress threads.

3.7.5 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.7.6 *Spring.*—The spring shall be of steel spring stock, heat-treated to give maximum efficiency, and shall produce immediate response in opening the jaws when the vise screw is loosened.

3.7.7 *Pivot link.*—The pivot link shall be of steel and securely attached to the back jaw. The hole for the pivot pin shall be parallel to the face of the back jaw and in alignment with the pivot pin hole in the arm of the front jaw. Means shall be provided for securing the pivot pin when inserted in the pivot link.

3.7.8 Type II vise shall conform to table II for the size specified (see 6.2) and shall be similar to figure 3.

TABLE II.—*Type II, blacksmith's*

Jaw width (nominal) ± 1/8 inch	Jaw opening (minimum)	Plate to bottom of body (approximate)	Weight (minimum)
Inches	*Inches*	*Inches*	*Pounds*
4	3¾	32	30
4½	4½	32	45
5	5¼	31	65
6	6	28	90
7	7	28	135
8	8	28	180

3.8 **Type III, sheetmetal worker's.**—Type III vise shall consist of a body, sliding member, swivel-base plate, vise screw, nut, and screw handle. The vise shall have high tapering jaws and shall be subjected to the test loads specified in 4.5.1.

3.8.1 *Body.*—The body shall be of cast iron, malleable iron, steel casting, or steel forg-

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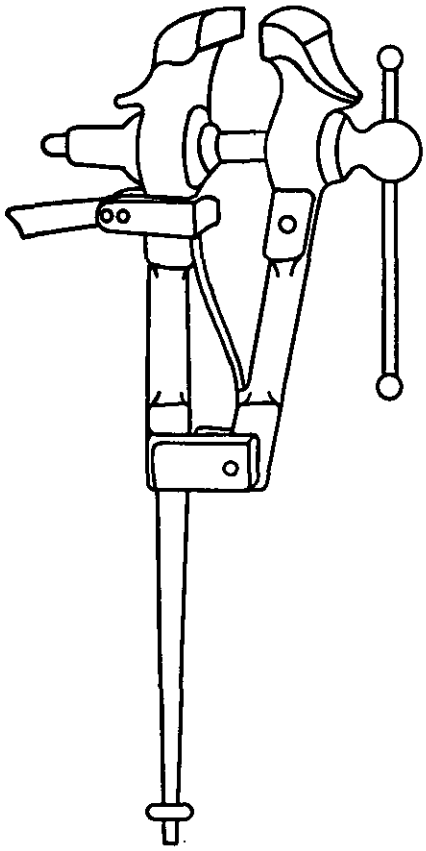


FIGURE 3.—Type II, blacksmith's.

ings. The back jaw shall be an integral part of the body. The plane of the inside face of the jaw shall be perpendicular with the axis of the hole in the body through which the sliding member passes, and shall distribute even pressure over the entire area of the face.

3.8.2 *Sliding member.*—The sliding members can be either rectangular or round. The front jaw, except for jaw facings, shall be of cast iron, malleable iron, or steel, or a combination of these materials, integral or attached to the sliding member. The sliding member shall have a smooth-machined or cold-rolled finish to allow not more than 0.012 inch at any point between the sliding member and its counterpart. The member shall slide in the body with smooth action without binding at any point throughout the full range of travel. Machined portions of sliding members shall be sufficiently rugged to withstand abrasion of filing, grindings, or impact from knocks to which they are normally subjected. Backlash in either direction shall not exceed one-eighth of one revolution of the handle. A collar or other means shall be provided to prevent the screw from coming out of the sliding member.

3.8.3 *Jaw facings.*—Jaw facings shall be of tool steel, or mild steel milled parallel with part or face that is applied to the castings, ground smooth, and properly carburized, and shall be heat-treated to a Rockwell hardness of from 53 to 59 on the C scale (see 4.5.2). The jaw facings shall be replaceable and shall be securely fastened to the jaws by machine screws. Unless otherwise specified (see 6.2), the work contact area shall be diamond or square serrated. All jaw facings shall have burrs removed and sharp edges broken.

3.8.4 *Swivel base plate.*—The swivel base plate shall be of the same materials as the body or steel forging. Provision shall be made for at least three bench mounting screws or bolts. The vise, when mounted on the plate, shall swivel through 360 degrees and shall have a positive locking arrangement to secure the vise in any angular position.

3.8.5 *Vise screw.*—The vise screw shall conform to the requirements specified in 3.6.3 except the screw shall have square, acme, or buttress threads.

3.8.5.1 *Nut.*—The nut, when not an integral part of the body, shall be of steel or malleable iron and conform to the requirements specified in 3.7.3.

3.8.6 *Screw handle.* — The screw handle shall conform to the requirements specified in 3.6.4.

3.8.7 Type III vise shall conform to table III and shall be similar to figure 4.

3.9 Type IV, bench and pipe (swivel base, stationary jaw).—Type IV vise shall consist of a body, sliding member, swivel-base plate,

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TABLE III.—*Type III, sheetmetal worker's*

Jaw width (nominal) ±1/16 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)			
					Cast iron		Steel forging, steel casting, malleable iron; steel cast iron [1]	
					Complete vise	Front-jaw sliding bar and screw	Complete vise	Front-jaw sliding bar and screw
Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
4½	9	⅞	1⅝	2⅜16	72	25	55	19

[1] "Steel cast iron" applies to vises with cast-iron bodies and steel sliding bars.

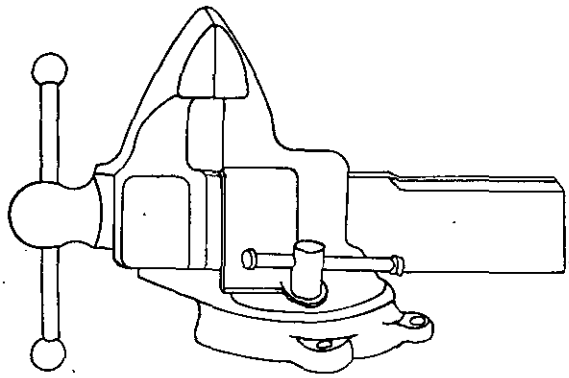


FIGURE 4.—Type III, sheetmetal worker's.

vise screw, nut, screw handle, and pipe jaws. The vise shall be subjected to the test load specified in 4.5.1.

3.9.1 *Body.*—The body shall conform to the requirements specified in 3.8.1.

3.9.2 *Sliding member.*—The sliding member shall conform to the requirements specified in 3.8.2.

3.9.3 *Swivel-base plate.*—The swivel-base plate shall conform to the requirements specified in 3.8.4.

3.9.4 *Vise screw.*—The vise screw shall conform to the requirements specified in 3.8.5.

3.9.5 *Nut.*—The nut, when not an integral part of the body, shall conform to the requirements specified in 3.7.3.

3.9.6 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.9.7 *Pipe jaws.*—The pipe jaws shall be of tool steel heat-treated to a Rockwell hardness value of not less than 52 nor more than 58 on the C scale (see 4.5.2). The jaw shall be located on the front jaw and body of the vise between the main jaw facings and the sliding member. The jaws, when attached to the front jaw and body, shall not project beyond the vertical line of the main jaw facings and shall be easily removed. The face of the jaws shall be V-shaped and have sharp V-serrations to grip the pipe rigidly.

3.9.8 *Jaw facings.*—The jaw facings for type IV vises shall conform to the requirements specified in 3.8.3 (see 6.2).

3.9.8.1 *Jaw caps.*—Jaw caps, when specified (see 6.2), shall conform to the requirements specified in 3.5.2.

3.9.8.2 Type IV vise shall conform to table IV for the size specified (see 6.2) and shall be similar to figure 5.

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TABLE IV.—*Type IV, bench and pipe (swivel base, stationary jaw)*

Jaw width (nominal) ±1/16 inch	Nominal pipe size capacity	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)			
						Cast iron		Steel forging, steel casting, malleable iron; steel cast iron [1]	
						Complete vise	Front-jaw sliding bar screw and pipe jaw	Complete vise	Front-jaw sliding bar screw and pipe jaw
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
3½	¼ to 2½	3	¾	1¼	1⅞	45	17	35	12
4½	¼ to 3½	5	⅞	1⅝	2⅜	75	26	55	19
5	¼ to 4½	6	1	2	2½	115	40	90	31
6	¼ to 6	8	1⅛	2	2¹³⁄₁₆	175	61	150	52

[1] "Steel cast iron" applies to vises with cast-iron bodies and steel sliding bars.

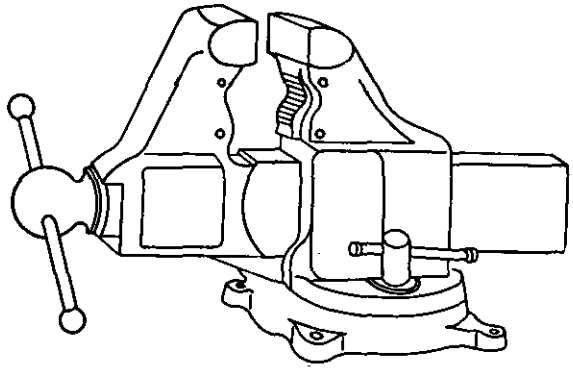


FIGURE 5.—Type IV, bench and pipe (swivel base, stationary jaw).

3.10 Type V, hand.

3.10.1 *Class 1, hinged jaw.*—The vise shall be either forged steel or semi-steel and consist essentially of two jaws hinged together at the opposite end from the clamping surfaces, a spring to hold the jaws in the open position, and an operating screw with wing nut and thrust washer to prevent the wing nut from cutting into the body.

3.10.1.1 *Jaws.*—The jaw facings shall have a Rockwell hardness of 50 to 56 on the C scale (see 4.5.2), and shall be shallow serrated.

3.10.1.2 *Spring.*—The jaws shall be distended by a steel spring securely contained between the jaws to prevent displacement.

3.10.1.3 *Screw.*—The screw shall have a standard thread and either a wing or knurled nut.

3.10.1.4 Class 1 vise shall conform to the dimensions shown in table V and shall be similar to figure 6.

3.10.2 *Class 2, hollow handle.*—Class 2 vise shall consist of two jaws and a body with handle.

3.10.2.1 *Jaws.*—Jaws shall be as specified in 3.10.1.1 and have a reamed hole for the insertion of the hinge pin, and the back of each jaw shall extend beyond the hinge to the taper on the handle.

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TABLE V.—*Type V, class 1, hand, hinged jaw*

Jaw width (nominal) ±1/16 inch	Jaw opening (minimum)	Over-all length	
		Minimum	Maximum
Inches	*Inch*	*Inches*	*Inches*
1½	1	4½	5½

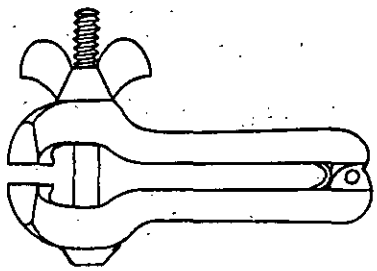


FIGURE 6.—Type V, class 1, hand, hinged jaw.

3.10.2.2 *Body and handle.*—The body and handle shall have a hole extending through both members of a size comparable to the capacity of the vise to permit the clamping of long rods and wire. The body shall be of steel threaded on one end to engage the threads in the handle. The other end shall have two opposed yoke sections to receive the vise jaws. The handle shall be of steel with a taper to actuate the jaws. The taper section of the handle shall have such a surface finish as to operate freely. The grip section of the handle shall be either straight or diamond knurled.

3.10.2.3 Class 2 vise shall conform to table VI for the size specified (see 6.2) and shall be similar to figure 7.

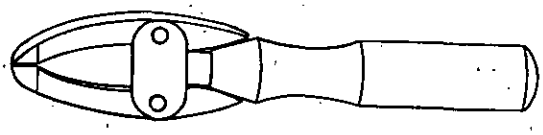


FIGURE 7.—Type V, class 2, hand, hollow handle.

3.10.3 *Class 3, combination bench and hand.*—Class 3 vise shall consist of a body, a pair of jaws, vise screw, screw handle, and vise handle, and shall be furnished with or without bench clamp.

3.10.3.1 *Body.*—The body shall be of steel and have two holes, one for the vise screw and the other for the jaw guide pin. The vise screw mounting hole shall be provided with a retaining arrangement for the screw and shall permit the screw to turn without binding. The lower end of the body shall be threaded to allow for mounting on the handle or bench clamp.

3.10.3.2 *Jaws.*—The jaws shall be of steel forgings and shall have a vise screw. The jaws shall be closed by a clockwise rotation of the screw. Guide pins or other arrangements shall be incorporated to maintain parallelism of the jaws. The facings of both jaws shall be shallow diamond serrated and have a Rockwell hardness of 50 to 56 on the C scale (see 4.5.2). One of the jaw facings shall have a 90-degree V-notch 1/8 inch wide, in the center.

TABLE VI.—*Type V, class 2, hand, hollow handle*

Jaw width ± 1/16 inch	Capacity jaw opening (minimum)	Over-all length (approximate)
Inch	*Inch*	*Inches*
¼	1/16	4½
¼	⅛	4½
½	¼	4½

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3.10.3.3 *Vise screw.*—The vise screw shall conform to the requirements specified in 3.6.3.

3.10.3.4 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.10.3.5 *Vise handle.*—The vise handle, if made of nonmetallic material, shall have a threaded steel insert for mounting to the body. When plastic handles are furnished, they shall be capable of withstanding the test specified in 4.5.4.

3.10.3.6 *Style A, without bench clamp.*—The vise shall be furnished without bench clamp.

3.10.3.7 *Style B, with bench clamp.*—The vise shall be furnished with a bench clamp of malleable iron or forged steel.

3.10.3.7.1 *Bench clamping screw.* — The bench clamping screw shall conform to the requirements specified in 3.6.5.

3.10.3.8 Class 3, style A and B vises shall conform to table VII and shall be similar to figure 8.

TABLE VII.—*Type V, class 3, hand, combination bench and hand*

Jaw width (nominal) ± 1/16 inch	Jaw opening (minimum)	Length over handle (approximate)
Inches	*Inches*	*Inches*
1½ ----------	1½	7

3.11 Type VI, jeweler's.—Type VI vise shall conform to table VIII for the size specified (see 6.2).

3.11.1 *Class 1, stationary base.*—Class 1 vise shall consist of a body, sliding member, vise screw, nut, and screw handle. The vise

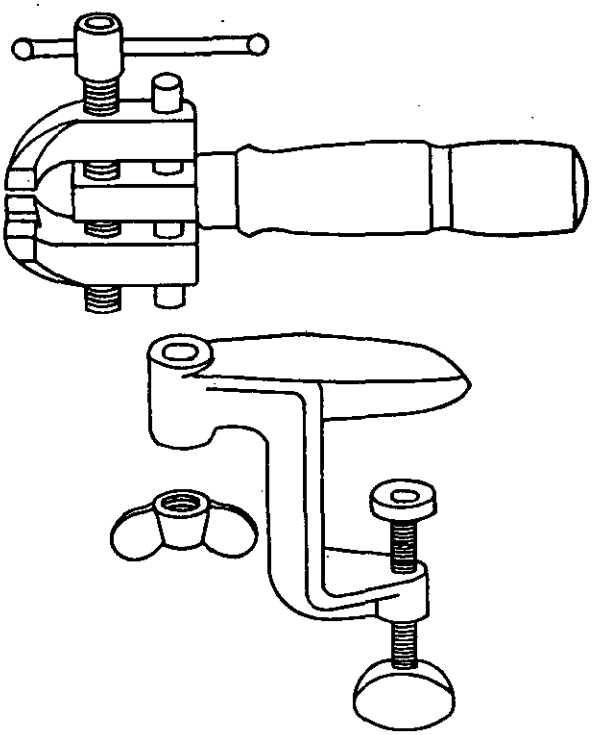


FIGURE 8.—Type V, class 3, hand, combination bench and hand, style A without bench clamp and style B with bench clamp.

shall be subjected to the test loads specified in 4.5.1.

3.11.1.1 *Body.* — The body shall have a flange or series of lugs around its lower edge and shall be provided with at least three vise mounting screws or boltholes. The body shall conform to the requirements specified in 3.8.1.

TABLE VIII.—*Type VI, jeweler's*

Jaw width (nominal) ± 1/16 inch	Jaw opening (minimum)	Screw diameter (minimum)	Weight (approximate)
Inches	*Inches*	*Inch*	*Pounds*
1¼ ----------	1	⅝	1
1½ ----------	1½	⅝	1½
2 ----------	2	⅝	4

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3.11.1.2 *Sliding member.* — The sliding member shall conform to the requirements specified in 3.8.2 except the sliding member shall be finished to allow not more than 0.008-inch clearance at any point between the sliding member and its counterpart.

3.11.1.3 *Vise screw.*—The vise screw shall conform to the requirements specified in 3.6.3.

3.11.1.4 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.11.1.5 *Jaw facings.*—The jaw facings for class 1 vise shall conform to the requirements specified in 3.8.3.

3.11.1.6 *Nut.*—The nut, when not an integral part of the body, shall conform to the requirements specified in 3.7.3.

3.11.1.7 Class 1 vise shall be similar to figure 9.

3.11.2 *Class 2, swivel base.*—Class 2 vise shall consist of a body, sliding member, swivel-base plate, vise screw, and screw handle. The vise shall be subjected to the test loads specified in 4.5.1. The vise shall conform to the requirements specified in 3.8.1 through 3.8.6, and shall be similar to figure 10.

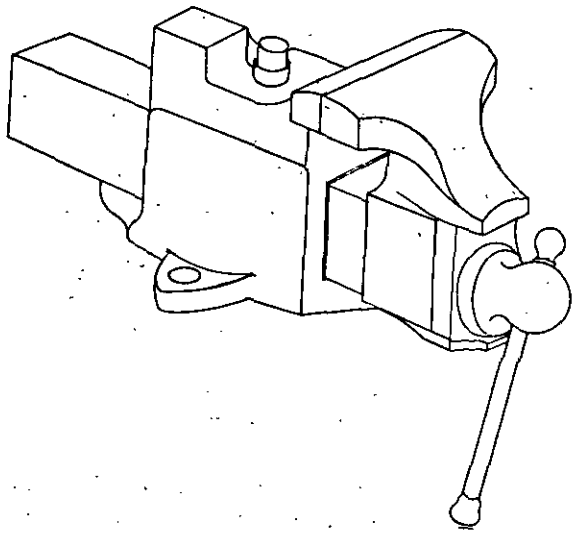


FIGURE 9.—Type VI, class 1, jeweler's, stationary base.

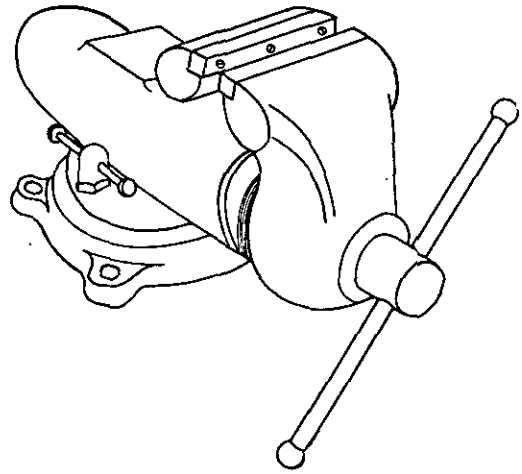


FIGURE 10.—Type VI, class 2, jeweler's, swivel base.

3.12 **Type VII, machinists.**—Type VII vise shall withstand the tests specified in 4.5.1.

3.12.1 *Class 1, stationary base, stationary jaw.*—Class 1 vise shall conform to the requirements specified in 3.8.1, 3.8.2, 3.8.3, 3.8.5, 3.8.5.1, and 3.8.6, and shall be provided with a stationary base.

3.12.1.1 *Stationary base.*—The body shall have a flange or series of lugs around its lower edge and shall be provided with at least three vise mounting screw or boltholes.

3.12.1.2 Class 1 vise shall conform to table IX and shall be similar to figure 11.

3.12.2 *Class 2, swivel base, stationary jaw.*—Class 2 vise shall conform to the requirements specified in 3.8.1 through 3.8.6, and table X for the size specified (see 6.2) and shall be similar to figure 12.

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TABLE IX.—*Type VII, class 1, machinist's, stationary base, stationary jaw*

Jaw width (nominal) ±1/8 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)			
					Cast iron		Steel forging, steel casting, malleable iron; steel cast iron [1]	
					Complete vise	Front-jaw sliding bar and screw	Complete vise	Front-jaw sliding bar and screw
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
2½	3	½	1	1¼	10	4	10	4
3	4	5/8	1¼	1 9/16	20	9	20	8
3½	5	¾	1½	1⅞	29	11	27	11
4	6	13/16	1⅝	2 3/16	38	17	36	13
4½	7	⅞	1⅝	2 3/16	52	22	47	19
5	8	1	1⅞	2½	73	29	57	24
6	9	1⅛	2	2 13/16	125	51	90	35
8	12	1¼	2 9/16	3 7/16	236	97	150	56

[1] "Steel cast iron" applies to cast-iron bodies and steel sliding members.

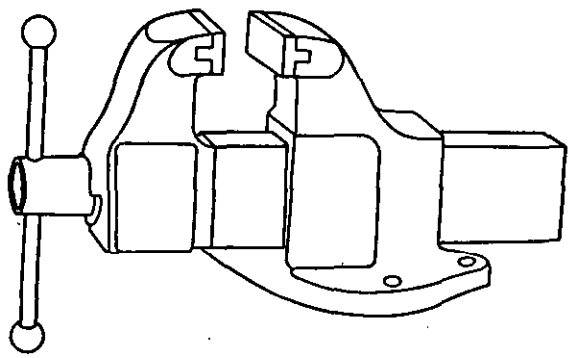


FIGURE 11.—Type VII, class 1, machinist's, stationary base, stationary jaw.

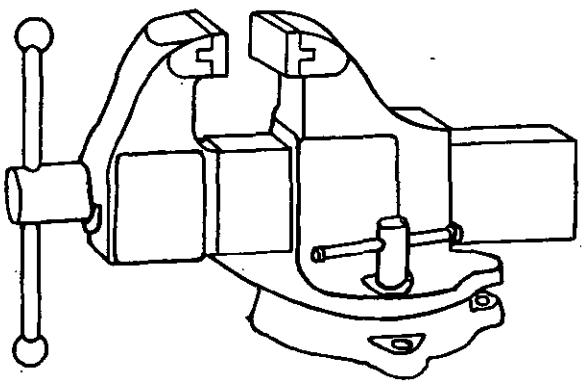


FIGURE 12.—Type VII, class 2, machinist's, swivel base, stationary jaw.

3.12.3 *Class 3, swivel base, swivel jaw.*— Class 3 vise shall conform to the requirements specified in 3.8.1 through 3.8.6, except that the vise shall be provided with a swivel back jaw.

3.12.3.1 *Swivel back jaw.*—The swivel back jaw shall be of the same material as the body and shall be held parallel to the front jaw by means of a steel taper pin. When the

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TABLE X.—*Type VII, class 2, machinist's, swivel base, stationary jaw*

Jaw width (nominal) ±⅛ inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)			
					Cast iron		Steel forging, steel casting, malleable iron; steel cast iron [1]	
					Complete vise	Front-jaw sliding bar and screw	Complete vise	Front-jaw sliding bar and screw
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
2½	3	½	1	1¼	14	5	13	4
3	4	⅝	1¼	1 9/16	27	9	25	8
3½	5	¾	1½	1⅞	34	11	34	12
4	6	13/16	1⅝	2 3/16	51	17	42	14
4½	6¾	⅞	1⅝	2 3/16	65	22	55	19
5	8	1	1⅞	2½	85	29	69	24
6	9	1⅛	2	2 13/16	150	51	100	35
8	12	1¼	2 3/16	3 7/16	278	97	170	57

[1] "Steel cast iron" applies to cast-iron bodies and steel sliding members.

steel pin is removed, the back jaw shall swivel free on a horizontal plane permitting the jaw to align with tapered work. Whether in the stationary or swiveled position, the vise shall clamp the work rigidly.

3.12.3.2 Class 3 vise shall conform to table XI for the size specified (see 6.2) and shall be similar to figure 13.

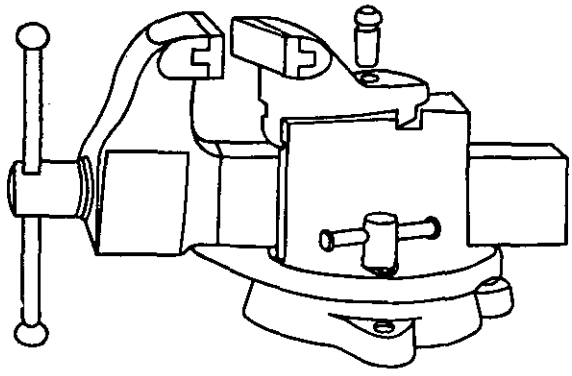


FIGURE 13.—Type VII, class 3, machinist's, swivel base, swivel jaw.

3.13 Type IX, pin.—Type IX vise shall consist of a chuck and handle. The handle shall afford a comfortable handgripping surface and be hollowed the full length to permit the work to pass through.

3.13.1 *Chuck.*—The chuck shall be of steel and consist of three or four jaws and a sleeve. The design of the chuck shall be such as to permit parallel movement of the gripping surfaces of the jaws and hold work concentrically of a diameter within the rated capacity of the vise.

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TABLE XI.—*Type VII, class 3, machinist's, swivel base, swivel jaw*

Jaw width (nominal) ±1/8 inch	Jaw opening (minimum)	Screw diameter (minimum)	Screw head diameter (minimum)	Nut threaded length (minimum)	Weight (minimum)			
					Cast iron		Steel forging, steel casting, malleable iron; steel cast iron [1]	
					Complete vise	Front-jaw sliding bar and screw	Complete vise	Front-jaw sliding bar and screw
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*	*Pounds*
3½	4	¾	1½	1⅜	36	11	35	11
4½	5½	⅞	1⅝	2⅜	71	23	56	19
6	9	1⅜	2	2¾	164	51	116	35

[1] "Steel cast iron" applies to cast-iron bodies and steel sliding members.

3.13.1.1 *Jaws.*—The jaws shall be of steel and show a Rockwell hardness value of not less than 48 nor more than 56 on the C scale (see 4.5.2), or the jaws may be casehardened to show a hardness value of not less than 84.5 nor more than 88.5 on the Rockwell 15N scale (superficial). The slots between the jaws shall extend a distance of not less than three-fourths the length of the chuck jaw to permit articulation of the jaw segments.

3.13.1.2 *Sleeve.*—The sleeve shall have a knurled grip for adjustment and shall be tapered at the nose end to provide work clearance. The sleeve shall have a metallic, oxide, or phosphate coating produced by chemical treatment of the steel.

3.13.2 *Class 1, metal handle.*—The outside diameter of the handle shall be knurled three-fourths of the exposed length advancing from the chuck end. The smooth portion of the handle shall be concentric with the axis of the vise in order that the vise may be held in machine chucks or collets. The handle shall have a metallic, oxide, or phosphate coating produced by chemical treatment of the steel.

3.13.2.1 Class 1 vise shall conform to table XII for the size specified (see 6.2) and shall be similar to figure 14.



FIGURE 14.—Type IX, class 1, pin, metal handle.

3.13.3 *Class 2, nonmetallic handle.*—The handle shall be of hardwood, plastic, or other nonmetallic material. When plastic handles are furnished, they shall be capable of withstanding the test specified in 4.5.4. The vises shall conform to table XIII and shall be similar to figure 15.



FIGURE 15.—Type IX, class 2, pin, nonmetallic handle.

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TABLE XII.—*Type IX, class 1, pin, metal handle*

Nominal capacity	Over-all diameter (maximum)	Over-all length	
		Minimum	Maximum
Inch	*Inch*	*Inches*	*Inches*
0.000 to 0.040	11/32	2¾	3¼
.025 to .062	13/32	3¼	3¾
.045 to .125	15/32	3½	4
.110 to .187	17/32	3¾	4¼
.175 to .250	19/32	3⅞	5

TABLE XIII.—*Type IX, class 2, pin, nonmetallic handle*

Nominal capacity
Inch
0 to 5/32
0 to ¼

3.14 Type X, pipe.

3.14.1 *Class 1, chain.*—Class 1 vise shall consist of a base, jaw or jaws, screw, chain, handle, and nut.

3.14.1.1 *Base.*—The base shall be of forged steel, malleable iron, or cast iron and have at least three bench mounting holes. The base shall be notched or other means shall be provided to anchor the chain by means of the chain links.

3.14.1.2 *Jaw or jaws.*—The jaw or jaws shall be of steel and either rigidly mounted or forged integral with the base. The clamping surface of the jaw or jaws shall be V-shaped or semi-circular and have mill cut V-shaped teeth for gripping the pipe. The jaw or jaws shall show a Rockwell hardness value of not less than 53 nor more than 59 on the C scale (see 4.5.2).

3.14.1.3 *Screw.*—The screw shall be in accordance with 3.5. One end of the screw shall be designed for attaching the chain with provisions for replacement, and the other end shall extend through a hole in the base and engage the nut.

3.14.1.4 *Chain.*—The chain shall be of steel having a tensile strength of not less than 90,000 p.s.i. (see 4.5.3), and be of the flat link type with projecting link pins to engage the slot in the base. The chain shall be replaceable and of such length as to grip the pipe of the maximum size for which the vise is designed.

3.14.1.5 *Handle.*—The handle shall be of forged steel or malleable iron or integral pin and yoke section of forged steel and secured to the nut.

3.14.1.6 *Nut.*—The nut shall be either malleable iron or of the same type of steel as the screw, threaded internally to engage the thread of the screw. The length of the nut shall be at least equal to the major diameter of the screw.

3.14.1.7 Class 1 vise shall conform to table XIV for the size specified (see 6.2) and shall be similar to figure 16.

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TABLE XIV.—*Type X, class 1, pipe, chain*

Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	*Inch*	*Pounds*
⅛ to 2½	11/16	3¾
¼ to 4	¾	10
⅜ to 6	⅞	18
½ to 8	⅞	30

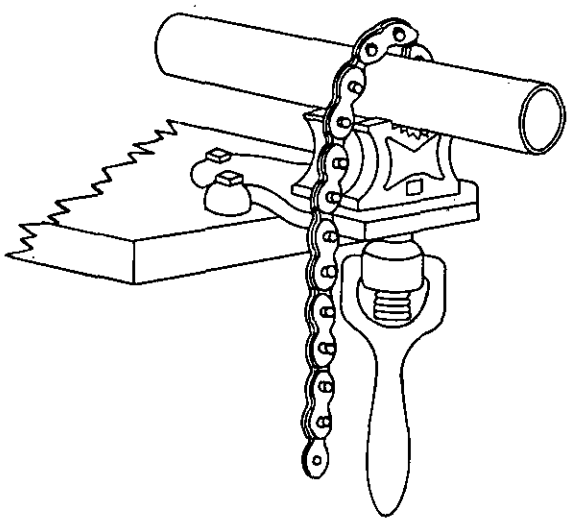


FIGURE 16.—Type X, class 1, pipe, chain.

3.14.2 *Class 2, hinged jaw.*—The vise shall consist of a base, frame and hook, jaw crosshead, jaws, screw, and screw handle.

3.14.2.1 *Base.*—The base shall be of malleable iron or steel and have at least three bench mounting holes. The vertical upright section of the base shall be provided with holes for the mounting of the frame, and shall have a lug on both sides for engagement of the latch hook. The upright section shall be symmetrical about the vertical centerline to provide for hinging the frame from either the right or left side. The lower jaw mounting shall be so designed as to allow the vise to clamp the minimum pipe-size capacity of the vise.

3.14.2.2 *Frame and hook.*—The frame and hook shall be of malleable iron or steel and shall be U-shaped. The open ends of the frame shall have mounting boltholes to align with the boltholes in the uprights of the base. The closed end shall have provision for an internal threaded hole in the center of the frame or a means to hold a replaceable nut to engage the vise screw. One of the legs shall have a hole for mounting the latch hook; the latch hook shall be of steel or malleable iron and shall be designed and mounted to automatically lock the frame in the closed position. The assembled frame shall be so designed as to be reversible in a vertical plane to the base to provide for hinging the frame either from the right or left side of the base. The inside edge of the frame shall have smooth and parallel ways free of burrs and fins to act as guides for the jaw crosshead and shall permit the crosshead to slide freely over the entire length of adjustment without binding.

3.14.2.3 *Jaw crosshead.*—The jaw crosshead shall be of malleable iron or steel. The design shall be such that the strength will be adequate for the purpose intended. The top of the crosshead shall provide means to allow the end of the screw to swivel and be designed to both raise and lower the crosshead by action of the screw.

3.14.2.4 *Jaws.*—The jaws shall be of tool steel heat-treated to a Rockwell hardness value of not less than 53 nor more than 59 on the C scale (see 4.5.2). The gripping surface shall be V or semicircular in shape and have mill cut V-shaped teeth. The lower jaw mounting shall be designed to allow the vise to clamp the minimum pipe-size capacity of the vise.

3.14.2.5 *Screw.*—The screw shall conform to the requirements specified in 3.6.3.

3.14.2.6 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.14.2.7 Class 2 vise shall conform to table XV for the size specified (see 6.2), and shall be similar to figure 17.

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TABLE XV.—*Type X, class 2, pipe, hinged jaw*

Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	*Inches*	*Pounds*
⅛ to 2½	⅞	10
⅛ to 4½	1	23
⅛ to 6	1⅛	35
1 to 8	1¼	75
1½ to 12	1⅜	160

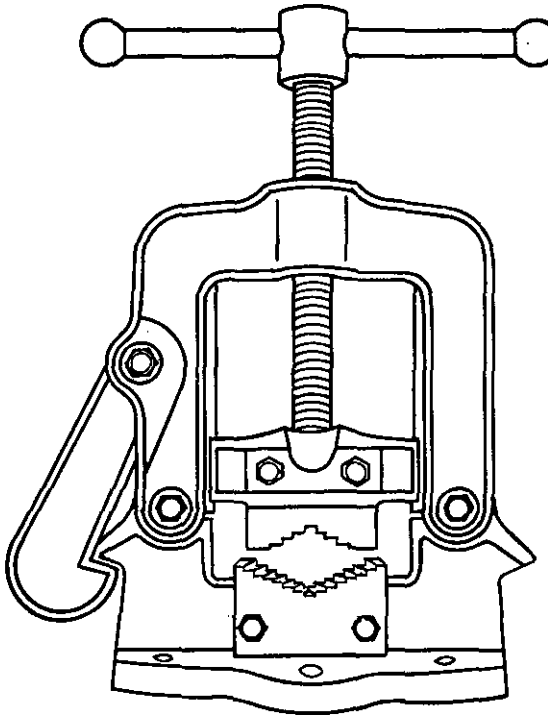


FIGURE 17.—Type X, class 2, pipe, hinged jaw.

3.14.3 *Class 3, open side.*—Class 3 vise shall consist of a base with column, jaw slide, jaws, screw, and screw handle.

3.14.3.1 *Base with column.*—The base shall be of malleable iron or steel with at least three bench mounting holes. The columns shall be an integral part of the base and have smooth and parallel guide ways free of burrs and fins or a guide bar exclusive of the column for the upper jaw slide, and have a threaded hole in the top portion of the column to engage the vise screw.

3.14.3.2 *Jaw slide.*—The jaw slide shall be of malleable iron or steel with provision for mounting the upper jaw and shall be guided through the entire capacity of the column without binding. One end of the screw shall swivel in the top section of the slide and the attachment shall be designed to allow for both raising and lowering the upper jaw by means of the vise screw.

3.14.3.3 *Jaws.*—The jaws shall be of tool steel and conform in detail and method of mounting as specified for class 2 vise (see 3.14.2.4).

3.14.3.4 *Screw.*—The screw shall conform to the requirements specified in 3.6.3.

3.14.3.5 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.14.3.6 Class 3 vise shall conform to table XVI for the size specified (see 6.2), and shall be similar to figure 18.

TABLE XVI.—*Type X, class 3, pipe, open side*

Nominal pipe size capacity	Screw diameter (minimum)	Weight (minimum)
Inches	*Inches*	*Pounds*
⅛ to 1½	¾	6½
⅛ to 2½	⅞	11½
⅛ to 4½	1⅛	27

3.15 Type XI, wire rope splicing.—Type XI vise shall consist of a body, three jaws, two screws, and two screw handles.

3.15.1 *Body.*—The body shall be of cast iron, malleable iron, or steel, having provision for at least three bench mounting screws or bolts, a horizontal jaw guide with a screw bearing at each end, and a vertical jaw guide with nut.

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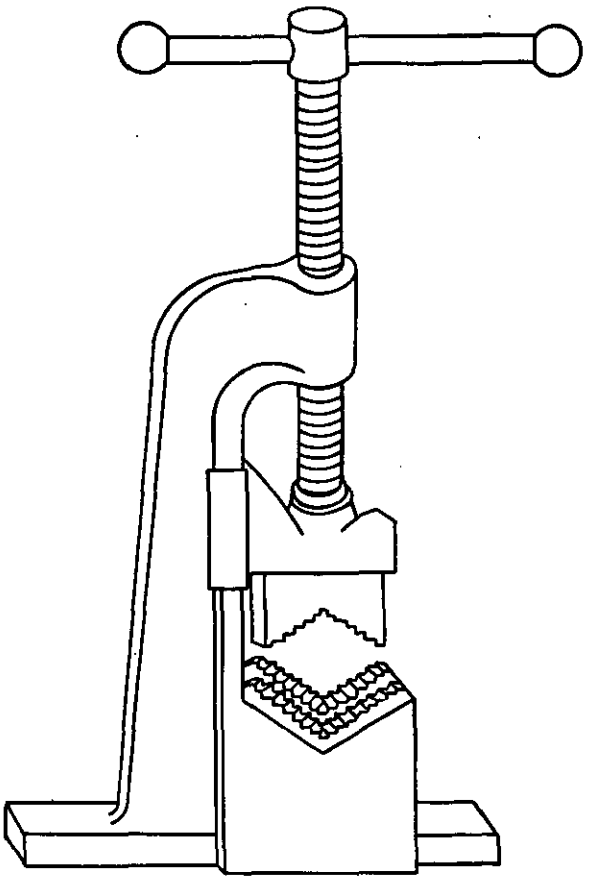


FIGURE 18.—Type X, class 3, pipe, open side.

3.15.2 *Jaws.*—The jaws shall be of cast steel or drop forged, designed to travel along the guides without binding for the entire distance of the vise capacity. When facing the vise, the nut portion of the left-hand jaw shall have right-hand threads, and the right-hand jaw, left-hand threads. The jaw faces shall have either one or two smooth grooves with all sharp edges broken and be capable of securely gripping the rope or cable around the thimble.

3.15.3 *Screws.* — Both the vertical and horizontal screws shall conform to the requirements specified in 3.5. One-half of the horizontal screw at handle end shall have left-hand threads; the other half of the screw shall have right-hand threads so as to activate the jaws in both opening and closing.

3.15.4 *Screw handles.*—The screw handles shall conform to the requirements specified in 3.6.4.

3.15.5 Type XI vise shall conform to table XVII for the size specified (see 6.2) and shall be similar to figure 19.

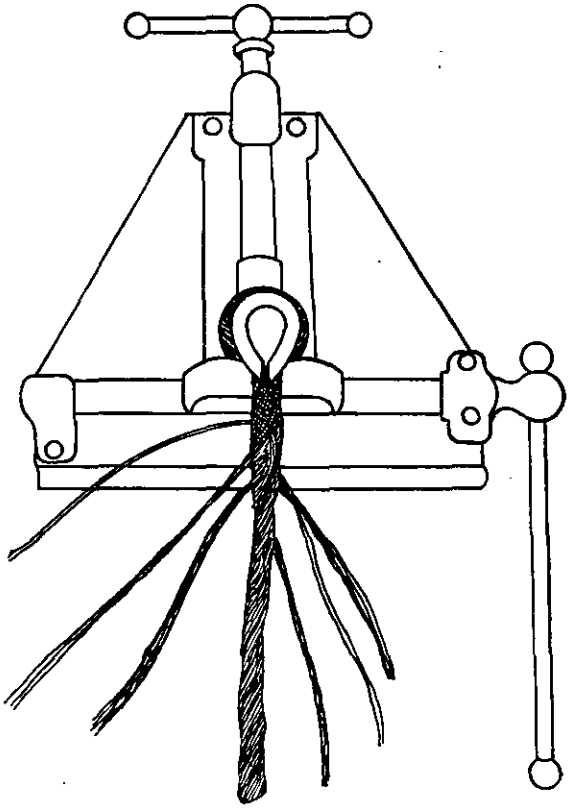


FIGURE 19.—Type XI, wire rope, splicing.

3.16 Type XII, handsaw filing.

3.16.1 *Class 1, with filing guide.*—Class 1 vise shall consist of a base, clamping screw, body with back jaw, front jaw, clamping lever, and filing guide.

3.16.1.1 *Base.*—The base shall be of gray or malleable iron casting and have a yoke section to form a bench clamping arrangement. A hole above the yoke section shall be provided for the body mounting.

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TABLE XVII.—*Type XI, wire rope splicing*

Nominal wire rope diameter capacity	Jaw width (minimum)		Jaw opening (minimum)	Screw diameter (minimum)		
	Front	Back		Left jaw	Right jaw	Vertical jaw
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
1/16 to 5/8	13/16	1 1/4	3 3/4	3/4	3/4	5/8
3/8 to 1 1/2	1 3/4	2	9 1/2	1 1/4	1 1/4	1 1/4
1 5/8 to 2 1/4	2	2 5/8	22	1 1/4	1 1/4	1 1/4

3.16.1.2 *Clamping screw.*—The clamping screw shall conform to the requirements specified in 3.6.5.

3.16.1.3 *Body with back jaw.*—The body shall be a gray or malleable iron casting and U-shaped. The depth and width of the U-section shall be such as to receive the widest and thickest of handsaws. The body shall be attached to the base in a manner as to permit the vise to be adjusted to various degrees of tilt. The adjustment shall be of such design as to rigidly hold the vise at any angular tilt. The back jaw shall be an integral part of the body and shall be located on the top of the back leg of the U-section. The front jaw shall pivot from the top of the front leg of the U-section by means of a hardened-steel pivot pin. The clamping lever shall be mounted to the front leg of the body casting in a manner similar to that of the front jaw.

3.16.1.4 *Front jaw.*—The front jaw shall be of the same material as the body with back jaw and shall have an extension to engage the cam or eccentric of the clamping lever.

3.16.1.5 *Clamping lever.* — The clamping lever shall be a gray or malleable-iron casting and shall close the jaws either by cam or eccentric action. The lever shall open the jaws when lifted and close them when lowered.

3.16.1.6 *Filing guide.*—The filing guide shall be mounted on the back jaw, shall be so designed as to guide the file at the proper angle, and shall be so mounted as to be detachable.

3.16.1.7 Class 1 vise shall conform to table XVIII for the size specified (see 6.2) and shall be similar to figure 20.

TABLE XVIII.—*Type XII, classes 1 and 2, handsaw filing, with or without filing guide*

Jaw width (nominal) ± ½ inch	Guide length (minimum)	Weight (minimum)
Inches	*Inches*	*Pounds*
9½	14½	7
12½	16	11

3.16.2 *Class 2, without filing guide.*—Class 2 vise shall conform to the requirements specified for class 1 (see 3.16.1) except that it shall not have a filing guide.

3.17 **Type XIII, band saw filing.**—Type XIII vise shall be of the same material and similar in design to type XII, class 1 (see 3.16.1) except that it shall have adjustable steps on both ends of the back jaw for locating the saw blades. Type XIII vise shall conform to table XIX for the size specified (see 6.2) and

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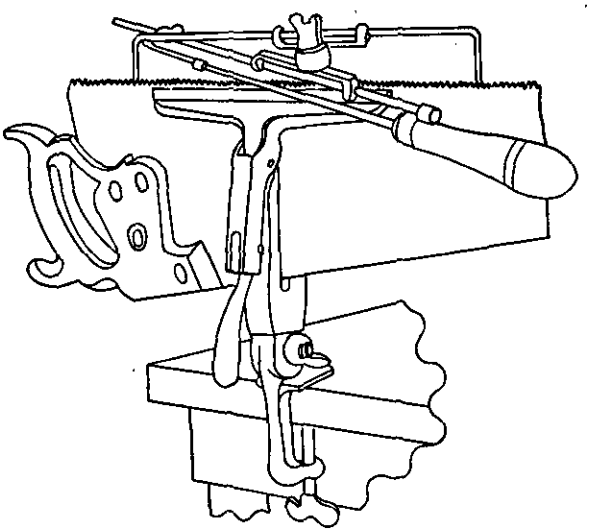


FIGURE 20.—Type XII, class 1, handsaw filing, with filing guide.

TABLE XIX.—*Type XIII, band saw filing*

Jaw width (nominal) ± ½ inch	Throat depth (minimum)	Weight (approximate)
Inches	*Inches*	*Pounds*
12¾	6	9½
36	7	60
42	10	100

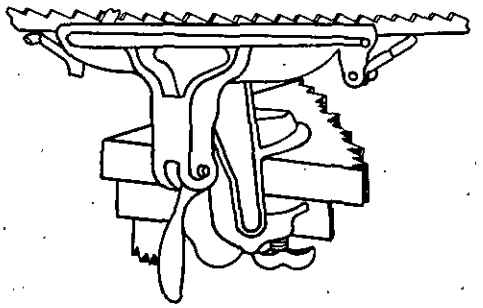


FIGURE 21.—Type XIII, band saw filing, jaw width under 15 inches.

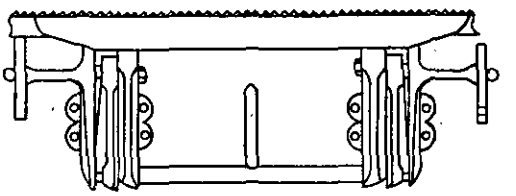


FIGURE 22.—Type XIII, band saw filing, jaw width over 15 inches.

shall be similar to figure 21 for jaw widths under 15 inches, and figure 22 for jaw widths over 15 inches.

3.18 **Type XIV, circular saw filing.**—Type XIV vise shall consist of a base, body with back jaw, front jaw, and vise screw with handle. The vise shall be capable of clamping circular saws from 5 to 18 inches in diameter.

3.18.1 *Base.*—The base shall be of gray, malleable, or cast iron. The base shall be provided with suitable means for bench mounting of the vise.

3.18.2 *Body with back jaw.*—The body shall be of the same material as the base. The back jaw shall be an integral part of the body. The plane of the inside face of the jaw shall be parallel with the inside face of the front jaw, and shall distribute even pressure over the entire area of the jaw face.

3.18.3 *Front jaw.*—The front jaw shall be of the same material as the body with back jaw. The front jaw shall be attached to the body in such a manner as to allow the jaw to open and close properly.

3.18.4 *Jaw facings.*—The jaw facings of both the back and front jaws which clamp the sides of the saw shall be properly hardened. The jaw facings shall be free from burrs and shall have all sharp edges broken.

3.18.5 *Vise screw.*—The vise screw shall be of commercial screwstock and in accordance with 3.5. The end of the screw shall be provided with means for receiving the handle.

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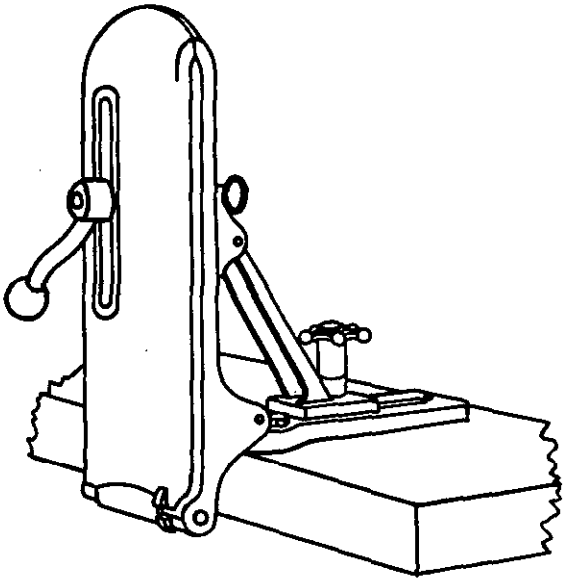


FIGURE 23.—Type XIV, circular saw filing.

3.18.6 *Screw handle.*—The screw handle shall be either a forged ball lever or a straight round steel bar of a diameter comparable to the size of the vise.

3.18.7 Type XIV vise shall be similar to figure 23 or 24.

3.19 Type XV, woodworker's.

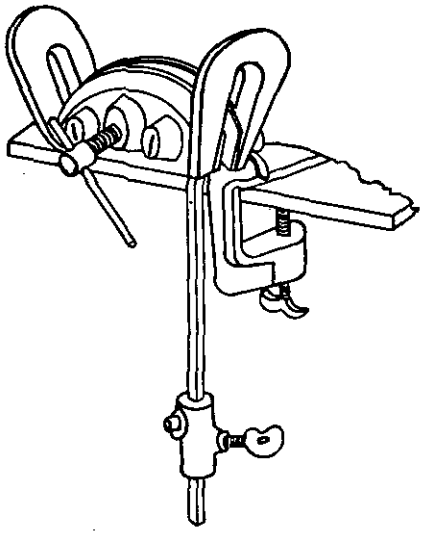


FIGURE 24.—Type XIV, circular saw filing.

3.19.1 *Class 1, continuous screw.*—Class 1 vise shall consist of a body with back jaw, front jaw, screw, nut, guide rods, screwhead, drag bar, and handle. The vise shall have a continuous-action screw and nut of either a single or double thread.

3.19.1.1 *Body.*—The body shall be of cast iron of solid construction, consisting of a back jaw, bench mounting, and rear bearing sections. The back-jaw section shall have at least two drilled holes in the upper portion of the jaw face for mounting of a wood facing. The lower portion of the back jaw shall have the front guide-rod bearings and the vise nut mounting. The bench mounting section shall be located on the rear side of the back-jaw section and shall be so machined on the top surface as to be square with the jaw face and shall have at least two bench mounting holes. The rear bearing section shall be so located and have bearing holes of such fit as to permit opening the jaws to their extreme capacity without bind.

3.19.1.2 *Front jaw.*—The front jaw shall be of cast iron and shall be permitted to incline a maximum of $\frac{1}{16}$ inch from vertical toward the top of the back of the jaw to compensate for any spring of the parts in clamping of work. The two guide rods shall be rigidly anchored to the bottom portion of the front jaw and the screw bearing shall be fastened between the guide rods. Four holes shall be provided for the mounting of a wood jaw facing. The front jaw shall be provided with at least one ½- by 1-inch brass or soft steel dog or stop.

3.19.1.3 *Screw.*—The screw shall be in accordance with 3.5, and have a shoulder or collar and bearing to fit the front jaw, and shall terminate in the screwhead on the front end and in a bearing in the drag bar on the other. The bearing surfaces on the screw shall be of such fit and the screwhead so located with reference to the shoulder or collar as to permit the screw to rotate freely.

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If the screw is of the continuous thread type, it may be either single or double threaded and shall be so designed that two revolutions of the double-thread screw or four complete revolutions of the single-thread screw shall move the front jaw a distance of 1 inch.

3.19.1.4 *Nut.*—The nut shall be of malleable iron, steel, or bronze, and be threaded to engage the screw, and shall be renewable.

3.19.1.5 *Guide rods.*—The guide rods shall be of cold-rolled steel. The axis of the rods shall be parallel to the axis of the screw, shall be securely fastened to the front jaw and extended through the holes in the body, and shall terminate in and be securely fastened to the drag bar. The minimum diameter of guide rods shall be ⅞ inch.

3.19.1.6 *Screwhead.*—The screwhead shall be of cast iron or malleable iron and shall be securely fastened to the screw. The screwhead shall be provided with a hole at right angles to the axis of the screw to receive the wooden, aluminum, or steel handle. The screwhead for the adjustable steel handles shall be provided with a tension spring to hold the handle at any desired location.

3.19.1.7 *Drag bar.*—The drag bar shall be of cast iron, malleable iron, or steel with holes drilled at both ends to fit the rear end of the guide rods. These holes shall be in alignment with those specified for the front jaw and body through which the guide rods are inserted. The hole in the center of the drag bar shall act as the rear bearing for the vise screw.

3.19.1.8 *Handle.*—The handle shall be of straight-grained hickory, solid aluminum, or steel. The handles shall be capped to prevent sliding out of the screwhead, or steel bar handles may have machine-type handle grips, securely mounted at right angles near each end.

3.19.1.9 Class I vise shall conform to table XX for the size specified (see 6.2) and shall be similar to figure 25.

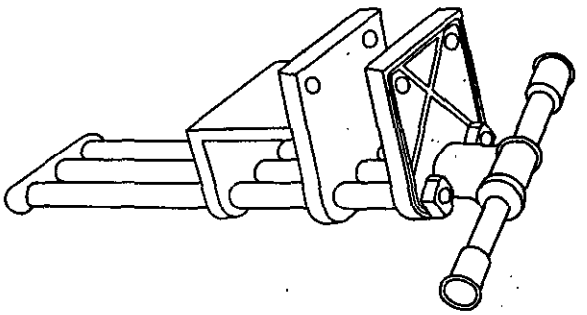


FIGURE 25.—Type XV, woodworker's, classes 1 and 2, continuous screw and quick acting.

3.19.2 *Class 2, quick acting.*—Class 2 vise shall be in accordance with class 1 vise (see 3.19.1) except it shall be provided with a device that will permit rapid adjustment of the jaw opening with the final pressure or release facilitated by the rotation of the screw in the conventional manner.

3.20 **Type XVI, patternmaker's.** — Type XVI vise shall consist of a body, square guide beam, front jaw, back jaw, tilt jaw, tilt support with vise lock, screw, and screw handle.

3.20.1 *Body.*—The body shall be of cast iron and consist of cast housing and locking lever. Housing shall be bored to permit working fit of back jaw boss, allowing full 360-degree rotation of jaws. Locking lever shall lock jaws at any angle of rotation.

3.20.2 *Square guide beam.* — The square guide beam shall be of cast iron and serve as a housing for the screw, and have a free sliding fit through the back jaw for the complete opening of the vise without binding. The square guide beam shall extend through both jaws and be securely fastened to the front jaw.

3.20.3 *Front jaw.*—The front jaw shall be of cast iron, rigid and parallel to the back jaw. The front jaw shall be capable of tilting 5 degrees in either direction about its vertical center line (or axle)—with a movement of a cam lever. Two holes shall be provided for the mounting of a soft facing jaw. The jaw shall have a horizontal machined groove for placement of the tilt jaw.

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TABLE XX.—*Type XV, woodworker's, classes 1 and 2, continuous screw and quick acting*

Jaw size (nominal)	Jaw dimensions (nominal)		Jaw opening (minimum)	Weight
	Width	Depth		
Inches	*Inches*	*Inches*	*Inches*	*Pounds*
7 by 4	7	4	9	20 to 35
10 by 4	10	4	12	30 to 45

3.20.4 *Back jaw.*—The back jaw shall be of cast iron, rigid and parallel to the front jaw. It shall have a large boss with machined opening permitting sliding fit of the beam. The outer surface of the boss shall be machined to permit 360-degree rotation of the back jaw beam and the front jaw in the body.

3.20.5 *Tilt jaw.*—The tilt jaw shall be of cast iron and shall be attached when needed to the front jaw for gripping of angular work. The jaw shall tilt 15 degrees up or down.

3.20.6 *Tilt support for vise with lock.*—The tilt support shall be of cast iron, and at least three holes shall be provided for mounting the vise to a bench top. It shall support the vise body on pivot pins to allow full 90-degree tilt of the vise from the normal horizontal position to a vertical position. A locking device shall permit the vise to be locked at any angle of tilt.

3.20.7 *Screw.*—The screw shall conform to the requirements specified in 3.6.3.

3.20.8 *Screw handle.*—The screw handle shall be of hardwood and capped at each end to preyent sliding out of the screwhead.

3.20.9 Type XVI vise shall conform to table XXI for the size specified (see 6.2) and shall be similar to figure 26.

3.21 Type XVII, machine table. — Type XVII vise shall consist essentially of a body with a fixed jaw, sliding jaw, screw, screw handle, and screw nut, and shall be suitable for machine or table mounting.

3.21.1 *Class 1, toolmaker's.*—Class 1 vise shall consist of a body with jaw and nut, sliding jaw, and screw handle.

3.21.1.1 *Body.*—The body shall be of low-carbon steel carburized and hardened to a Rockwell hardness of 58 to 65 on the C scale (see 4.5.2). One of the vise jaws and the nut shall be an integral part of the body. The nut portion of the body shall be threaded to receive the vise screw and have a threaded length of at least two and one-half times the diameter of the screw. The two sides shall be parallel to each other, and square with the inside of the slide surface bottom and back of body jaw and to the inside surface of body jaw with 0.001 inch.

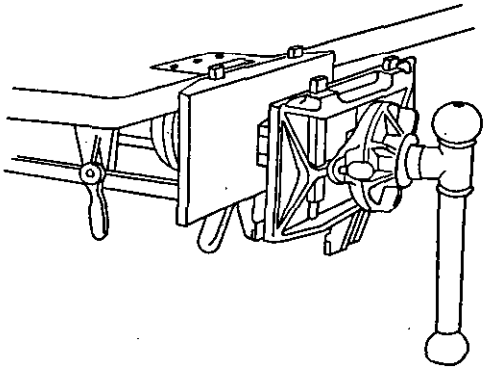


FIGURE 26.—Type XVI, patternmaker's.

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TABLE XXI.—*Type XVI, patternmaker's*

Jaw size (nominal)	Jaw dimensions		Jaw opening (minimum)	Weight (minimum)
	Depth ±½	Width ±1½		
Inches	*Inches*	*Inches*	*Inches*	*Pounds*
5 by 14	5	14	12	55
7 by 18	7	18	14	85

3.21.1.1.1 *Sliding jaw.*—The sliding jaw shall be of the same material and given the same heat treatment as the body. Each vise shall be furnished with one thick and one thin jaw. The swivel end of the screw and sliding jaw attachment shall be designed to permit interchangeability of the jaws and provide for the use of the screw in both opening and closing of the jaws.

3.21.1.1.2 *Screw with handle.*—The screw shall be of the best grade of steel screwstock with national extra-fine threads and shall be in accordance with 3.5. One end of the screw shall have a diamond knurled head with a steel bar handle securely fastened in place; the other end shall extend through the body nut and swivel in the interchangeable jaw. The threads of the screw shall conform to 3.5.1.

3.21.1.2 *Style A, slotted base.* — Style A body shall have a slot running parallel to the sides from the nut to the body jaw for key alignment of the screw jaw.

3.21.1.3 *Style B, solid base.*—Style B body shall be solid without key alignment slot and shall be subjected to the test loads specified in 4.5.1.

3.21.1.4 Class 1 vise shall conform to table XXII and shall be similar to figure 27.

3.21.2 *Class 2, drill.*

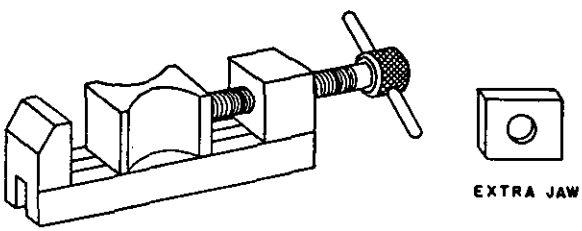


FIGURE 27.—Type XVII, class 1, styles A and B, machine table, toolmaker's, slotted base or solid base.

3.21.2.1 *Style A, quick action.*—Style A vise shall consist of a body with fixed jaw, sliding jaw, handle screw, swivel button, and nut.

TABLE XXII.—*Type XVII, class 1, machine table, toolmaker's*

Jaw width (nominal) ± 1/16 inch	Thin jaw opening (minimum)	Thick jaw opening (minimum)	Jaw depth (minimum)
Inch	*Inches*	*Inch*	*Inch*
⅞	2	1	¾

3.21.2.1.1 *Body with fixed jaw.*—The fixed jaw shall be an integral part of the body and both members shall be of cast iron or steel. The jaw facing shall be made of tool steel and be readily replaceable. The bottom surface shall be parallel with the sliding jaw surface, and square with the sides and fixed jaw

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clamping surface with 0.0002-inch full indicator reading. The tolerance as applied to the fixed jaw face and bottom shall be in the toe-in direction only. The body member shall be of such thickness as not to distort more than 0.001 inch when work is clamped hand tight at the maximum capacity of the vise. The body shall have a slot with serrations or taper for the attachment of the nut to provide for quick action in opening and closing the vise jaws. There shall be a flange on both ends of the body to permit the use of toe clamps or other arrangements in fastening the vise to machine tables.

3.21.2.1.2 *Sliding jaw.*—The sliding jaw shall be of the same material as the body and shall be machine finished on the bottom to slide on the slide surface of the body and be square with the jaw face within 0.002 inch. This tolerance shall be in the toe-in direction. A rectangular hole back of the jaw shall be provided for the vise nut.

3.21.2.1.3 *Handle screw and swivel button.*—The handle screw and swivel button shall be of steel, threaded on one end to engage the thread of the nut. The threaded end shall be provided with a swivel button to contact the sliding jaw; the other end shall be diamond knurled with a hexagonal section between the threaded-and-knurled portion to provide for wrench tightening of the vise jaws. The screw threads shall conform to 3.5.1.

3.21.2.1.4 *Nut.*—The nut shall be drop-forged steel designed to fit the rectangular hole in the sliding jaw member and shall have a projection to act as a key in the slot of the body. The quick action device of the nut shall be so designed that when the handle is released the sliding jaw will slide freely through the entire capacity of the vise and when the handle is lowered the nut shall be locked to the body member by means of cam arrangement in conjunction with a tapered key or serrations provided in the body and shall securely lock the nut to the body in any increment of the vise capacity.

3.21.2.1.5 Style A vise shall conform to table XXIII for the size specified (see 6.2) and shall be similar to figure 28.

TABLE XXIII.—*Type XVII, class 2, style A, machine table, drill, quick action*

Width of jaw ± 1/16 inch	Depth of jaw (minimum)	Jaw opening (minimum)	Weight (approximate)
Inches	*Inches*	*Inches*	*Pounds*
2	1 5/16	1 3/4	4 1/2
2 3/4	1 3/16	2 1/2	8 1/2
3 1/2	1 7/16	3	16

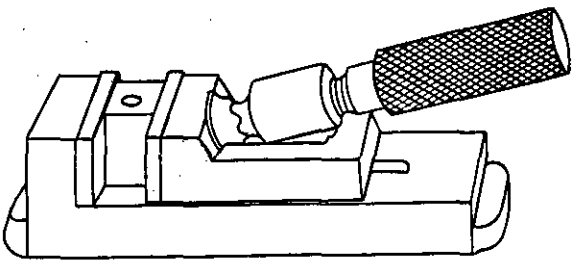


FIGURE 28.—Type XVII, class 2, style A, machine table, drill, quick action.

3.21.2.2 *Style B, continuous screw, light duty.*—Style B vise shall consist of a body with fixed jaw, sliding jaw, screw, screw handle, and screw nut.

3.21.2.2.1 *Body.*—The body, fixed jaw, and screw flange shall be integral, of cast iron or steel, and shall have at least three screw or bolt slots for mounting. The base of the body shall be slotted or grooved to form a guide for the sliding jaw. The face of the back jaw and the base shall be finished square and perpendicular to each other. The screw flange shall be opposite the back jaw and shall provide a rigid support for the screw nut.

3.21.2.2.2 *Sliding jaw.*—The sliding jaw shall conform in shape, dimension, and material to the back jaw. A projection to engage the base guide and to receive the end of

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the screw shall be incorporated in the jaw. The face of the jaw shall contain a central vertical 90-degree V-notch ¼ inch wide. The projection shall make a close sliding fit with the base guide. The face of the sliding jaw shall remain parallel to the face of the back jaw in any position of travel. Provision shall be made to prevent disengagement of the projection from the guide while the vise is in operation.

3.21.2.2.3 *Screw.* — The screw shall conform to 3.5. Screw action shall be continuous in both directions with no resistance except that of the sliding jaw. The screw shall be in constant alignment with the nut. One end of the screw shall engage the bushing on the sliding jaw, and the other shall project beyond the screw flange and terminate in an integral head.

3.21.2.2.4 *Screw handle.*—The screw handle shall be either a straight steel bar or a removable crank. The steel bar handle shall conform to the requirements of 3.6.4. When a removable crank is furnished, the crank shall be of forged steel or malleable iron and attached to the integral head of the screw.

3.21.2.2.5 *Screw nut.*—The nut shall be of malleable iron or steel and threaded to engage the threads of the screw. The nut shall be replaceable and held rigidly by the screw flange in alignment with the screw receiving projection in the sliding jaw without the use of shims. The nut shall be held against its shoulder or a shoulder in the screw flange by a socket head setscrew of suitable size. The threaded section of the nut shall completely enclose the screw threads.

3.21.2.2.6 Style B vise shall conform to table XXIV, and shall be similar to figure 29.

3.21.2.3 *Style C, continuous screw, heavy duty.*—Style C vise shall consist of a body with fixed jaw, sliding jaw, screw, screw nut, screw handle, and jaw facings.

3.21.2.3.1 *Body.* — The body shall be as specified in 3.21.2.2.1, except the top of the work base shall be provided with hardened-steel plates to prevent holes being drilled into the vise.

TABLE XXIV.—*Type XVII, class 2, style B, machine table, drill, continuous screw, light duty*

Width of jaw ± 1/16 inch	Depth of jaw (minimum)	Jaw opening (minimum)	Weight (approximate)
Inches	*Inches*	*Inches*	*Pounds*
3½ ----------	1¾	4	16

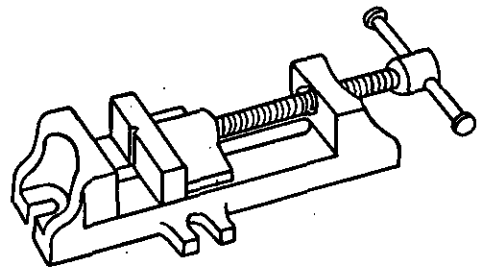


FIGURE 29.—Type XVII, class 2, style B, machine table, drill, continuous screw, light duty.

3.21.2.3.2 *Sliding jaw.*—The sliding jaw shall conform in shape, dimensions, and material to the back jaw. A projection to engage the base guide and receive the end of the screw shall be incorporated in the jaw. The face of the sliding jaw shall remain parallel to the face of the back jaw in any position of travel. Provision shall be made to prevent disengagement of the projection from the base guide while the vise is in operation.

3.21.2.3.3 *Screw.*—The screw shall conform to 3.6.3.

3.21.2.3.4 *Screw nut.*—The screw nut shall be of malleable iron or steel and threaded to engage the threads of the screw. The nut shall be replaceable and held rigidly by the screw flange in alignment with the screw receiving projection in the sliding jaw without

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the use of shims. The nut shall be held against its shoulder or a shoulder in the screw flange by a socket head setscrew of suitable size. The threaded section of the nut shall completely enclose the screw threads.

3.21.2.3.5 *Screw handle.*—The screw handle shall conform to 3.6.4.

3.21.2.3.6 *Jaw facings.*—The jaw facings shall conform to 3.8.3 (see 6.2).

3.21.2.3.7 Style C vise shall conform to table XXV and shall be similar to figure 30.

TABLE XXV.—*Type XVII, class 2, style C, machine table, drill, continuous screw, heavy duty*

Width of jaw ± 1/16 inch	Depth of jaw (minimum)	Jaw opening (minimum)	Weight (approximate)
Inches	*Inches*	*Inches*	*Pounds*
6	2	6	50

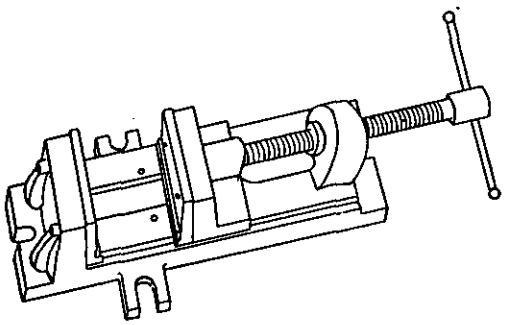


FIGURE 30.—Type XVII, class 2, style C, machine table, drill, continuous screw, heavy duty.

3.22 Type XVIII, bench and machine table.—Type XVIII vise shall consist of a body with jaw and nut, sliding jaw, swivel base, vise screw, jaw facings, and screw handle. The vise shall be designed to permit removing the vise from the swivel base without disturbing any work that may be clamped between the jaws.

3.22.1 *Body with jaw and nut.*—The body shall be of cast iron or steel. The jaw and nut shall be integral with the body. The bottom surface of the body shall be slotted longitudinally to allow the body to slide on the swivel base. The two sides of the body shall be parallel to each other, and square with the top and bottom slide surfaces and outside (or back) of body jaw within 0.002 inch. The nut portion of the body shall be threaded to receive the vise screw and shall have a threaded length of at least one and one-half times the diameter of the screw.

3.22.2 *Sliding jaw.*—The sliding jaw shall be of the same material as the body and shall be machine finished on the bottom surface and be square with the clamping surface of the jaw within 0.001 inch. The back of the jaw shall be provided with a swivel to permit opening and closing of the jaw by the vise screw.

3.22.3 *Swivel base.*—The swivel base shall be of cast iron, malleable iron, or steel; shall swivel through 360 degrees; and shall have a quick-acting locking device to rigidly lock the vise in any angular increment. The bottom of the swivel base shall be provided with a flange with a minimum of three holes for bench mounting screws or bolts.

3.22.4 *Vise screw.*—The vise screw shall have acme threads of a size in proportion with the size of vise on which it is used and shall be in accordance with 3.5. One end shall be headed and provided with a hole to receive the vise handle; the other end shall extend through the body nut and swivel in the sliding jaw. The threads of the screw shall conform to 3.5.1.

3.22.5 *Jaw facings.*—Each jaw shall be equipped with a renewable, steel jaw facing having a Rockwell hardness of 50 to 56 on the C scale (see 4.5.2). Each vise shall be furnished with a grooved steel insert for clamping of round or odd shaped work. The grooves shall be 90-degree V-shaped, six in number, three horizontal and three vertical.

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The grooves shall vary in width from 1/16 to 3/8 inch inclusive.

3.22.6 *Screw handle.*—The screw handle shall conform to the requirements specified in 3.6.4.

3.22.7 Type XVIII vise shall conform to table XXVI for the size specified (see 6.2) and shall be similar to figure 31.

TABLE XXVI.—*Type XVIII, bench and machine table*

Jaw width (nominal) ± 1/16 inch	Jaw opening (minimum)	Jaw depth (minimum)
Inches	*Inches*	*Inches*
1½	1½	⅞
2	2	1¼
2¾	2¾	1⅞
4	4	2⅛

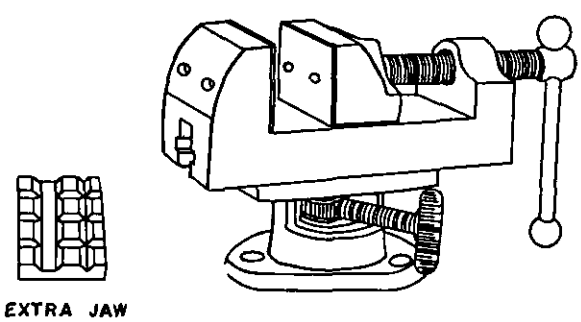


FIGURE 31.—Type XVIII, bench and machine table.

3.23 **Type XIX, contour gripping.**—Type XIX vise shall consist essentially of a body, front and back finger housings, front and back gripping fingers, locking levers, handle and handle screw. The vise shall be capable of securely gripping and holding square, round, tapered, or irregular-shaped objects when used either on the bench, in the tool room, or in machine work. The vise shall be subjected to the test loads specified in 4.5.1.

3.23.1 *Body.*—The body shall be of cast iron, malleable iron, steel castings, or steel forging. The plane of the bottom surface shall be parallel with the slide rails or ways within 0.003 inch. The front and back bearing holes for the handle screw shall be parallel with the slide rails or ways and be centrally located in the body to provide for proper alignment with the threaded hole in the front finger housing. The body shall be provided with at least four mounting screw or bolt slots around its lower edge.

3.23.2 *Front and back finger housings.*—The front and back finger housings shall be of cast iron, malleable iron, or steel, or combination thereof. The rail bearing surfaces shall have a smooth machined surface to allow not more than 0.012 inch at any point throughout the full range of travel. The machined portion of the rail bearing surface shall be sufficiently rugged to withstand abrasion of filings, ground particles, or impact from knocks to which they are normally subjected. The front finger housings shall be provided with a threaded lug or projection to engage the handle screw. The threaded portion of the lugs or projections shall be in alignment with the front and back bearing holes of the body to alleviate any tendency of the screw to bind. The front housing lug or projection shall have right-hand threads corresponding with the threads of the handle screw.

3.23.3 *Front and back gripping fingers.*—The front and back gripping fingers shall be of steel properly hardened and tempered to a Rockwell hardness value of not less than 55 nor more than 61 on the C scale (see 4.5.2). Unless otherwise specified (see 6.2), the work contact area of the gripping fingers shall be diamond or square serrated. Each individual gripping finger shall work independently of each other, sliding toward or away from the opposing finger. The front and back gripping fingers shall be constructed in such a manner that the gripping pressure is evenly distributed along the points of contact of the work piece.

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3.23.4 *Locking levers.* — Locking levers shall be provided for locking both the front and back gripping fingers. The locking levers shall be capable of locking the individual fingers in a position formed by the object held in order that an identical object may be clamped in the same manner without resetting the fingers. The opening and closing of the vise shall not distort the position of the locked gripping fingers. The locking levers, when in a neutral position, shall allow the gripping fingers to move freely without interference. The locking levers shall be of malleable iron or steel in order to withstand hard usage.

3.23.5 *Handles.*—The handles shall be of malleable iron or steel and shall have sufficient throw to readily tighten or loosen the handle screw with a minimum of effort. The handle shall have either a square or hexagon shaped opening for engaging the handle screw.

3.23.6 *Handle screw.*—The handle screw shall have either square, acme, or buttress threads and shall be in accordance with 3.5. The screw shall have right-hand threads to actuate the front finger housings in both opening and closing. The handle end of the screw shall be either square or hexagon shaped for engaging the handle.

3.23.7 Type XIX vise shall conform to table XXVII for the size specified (see 6.2), and shall be similar to figure 32.

3.24 **Workmanship.** — The workmanship shall be in accordance with the best grade of commercial practice covering this class of equipment. All parts shall be free from defects or blemishes that may affect the serviceability or appearance.

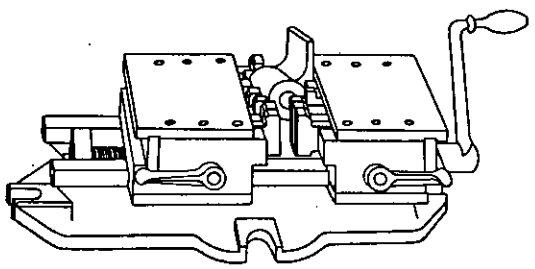


FIGURE 32.—Type XIX, contour gripping.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 **Inspection lot.**—All vises of the same type, class, style, and size offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2 **Sampling for lot acceptance.**

4.2.1 *Sampling for visual and dimensional examination.* — A random sample of vises shall be selected from each inspection lot of material offered for examination of visual and dimensional characteristics in accordance with Military Standard MIL–STD–105 at inspection level II for lots containing 180 items or less and inspection level I for lots containing more than 180 items. The acceptable quality level shall be 2.5 percent defective.

4.2.2 *Sampling for lot acceptance tests.*

4.2.2.1 *All tests except tension and flammability (when applicable).* — A random sample of vises shall be selected from each inspection lot offered for testing in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–8. The acceptable quality level shall be 4.0 percent defective. However, the sample size shall be the number associated with the letter in the table of the appendix and the acceptance number shall be zero until the acceptance number for the acceptable quality level is one or greater.

4.2.2.2 *Tension and flammability tests (when applicable).*—A random sample of vises shall be selected from each inspection lot offered for testing in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–3.

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TABLE XXVII.—*Type XIX, contour gripping*

Nominal size	Jaw width		Jaw opening (minimum)	Jaw depth (minimum)	Over-all length (approximate)	Over-all width (approximate)	Over-all height (approximate)	Weight (approximate)
	Minimum	Maximum						
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*
1½	1½	1⅝	1 17/32	¾	7⅛	3	2½	5
2	2	2⅛	3/16	1⅜	9¼	4¾	4	20
4	4	4⅛	3/16	2¼	16¼	7¾	5¼	75
6	6	6¼	6/16	3¾	21⅞	9¾	6⅝	120

4.3 Visual and dimensional examination.—Each of the sample vises selected in accordance with 4.2.1 shall be visually and dimensionally examined to verify compliance with this specification. Any vise in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective vises in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.1 *Back jaw face inspection, type I.*—Each sample type I vise selected in accordance with 4.2.1 shall be examined by the inspector to determine conformance with 3.6.1 as follows: The vise shall be set on a surface plate or true plane, with a straight edge against the face of the back jaw, and the point of contact of the straight edge with the surface plate shall be established. A right-angle square shall then be placed against the top edge of the back jaw, and the point of contact of the vertical leg of the square with the surface plate shall be established. The distance between these two points shall be measured. This distance shall not exceed 1 degree and may be in either direction from perpendicular.

4.4 Lot acceptance tests.

4.4.1 *All tests except tension and flammability.*—Each of the sample vises selected in accordance with 4.2.2.1 shall be subjected to the applicable tests specified in 4.5.1, 4.5.2, and 4.5.4.1 to verify compliance with this specification. Any sample vise which does not conform to these characteristics shall be rejected, and if the number of nonconforming vises in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 *Tension and flammability tests.*—Each of the sample vises selected in accordance with 4.2.2.2 shall be subjected to the applicable tests specified in 4.5.3 and 4.5.4.2 to verify compliance with this specification. If any vise in the sample fails either test, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 *Handle-loading test.*—Types III, IV, VII, and XIX vises shall be subjected to handle-loading tests to determine conformance with the handle-loading requirements of table XXVIII. Type VI and type XVII, class 1, style B vises shall be subjected to a handle-loading test of 1300 inch-pounds. These tests shall consist of loading the handle as near the end as practicable while the jaws clamp a parallel block of such dimensions that the jaw opening is equal to the nominal width of the jaws. The test load shall be determined by dividing the required turning moment by a perpendicular distance in

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inches from the axis of the screw to the line in which the load acts. The handle shall be considered as having taken a permanent set when the maximum deviation of the bent portion from a straight edge held against the longer straight portion is 1/32 inch or more. After the loading test the vise shall operate smoothly and the component parts shall show no distortion visible to the naked eye.

TABLE XXVIII.—*Handle loads*

Jaw width (nominal)	Test loads
Inches	*Inch-pounds*
1½	500
2	680
2½	800
3	1,300
3½	1,600
4	2,000
4½	2,700
5	3,700
6	7,000
8	15,000

4.5.2 *Hardness test.*—Jaws, jaw facings and other portions of each sample vise taken in accordance with 4.2.2.1 for which the applicable hardness requirements are specified shall be tested on a Rockwell tester. The readings shall be taken on the C scale using a diamond penetrator and employing a 150-kilogram load as specified in 3.8.3, 3.9.7, 3.9.8, 3.10.1.1, 3.10.2.1, 3.10.3.2, 3.11.1.5, 3.11.2, 3.12.1, 3.12.2, 3.12.3, 3.13.1.1, 3.14.1.2, 3.14.2.4, 3.14.3.3, 3.21.1.1, 3.21.2.3.6, 3.22.5, and 3.23.3. All hardness readings shall be made after removing the outer surfaces to a depth of 1/64 inch. Grinding of specimens in preparation for hardness tests shall be done by such means and in such a manner as to avoid tempering.

4.5.3 *Tension test.*—Each vise selected in accordance with 4.2.2.2 for which the applicable physical requirements of the steel are specified in 3.5, 3.6.3, 3.7.4, 3.8.5, 3.9.4, 3.10.3.3, 3.11.1.3, 3.12.1, 3.12.2, 3.12.3, 3.14.1.3, 3.14.1.4, 3.14.2.5, 3.14.3.4, 3.15.3, 3.18.5, 3.19.1.3, 3.19.2, 3.20.7, 3.21.1.1.2, 3.21.2.2.3, 3.21.2.3.3, 3.22.4, and 3.23.6 shall be tested in accordance with method 211 of Test Method Standard No. 151.

4.5.4 *Test procedures for plastic handles.*

4.5.4.1 *Heat distortion test.*—The handle of the vise for test shall be completely immersed in boiling water at 212° F. for a period of 2 minutes. At the end of this period, the sample shall be removed from the water and shall show no significant signs of distortion or blistering.

4.5.4.2 *Flammability test.*—The metal portion of the vise for test shall be clamped in a support with the handle extended outward in a horizontal position. Under the handle there shall be clamped a piece of 10-mesh Bunsen burner gauze about 5 inches square in a horizontal position, ¼ inch below the bottom of the handle, and with about ½ inch of the handle extending beyond the edge of the gauze. A Bunsen burner with an efficient flame ½ to ¾ inch in height shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall not exceed 2 inches per minute and shall be determined from the time required for the flame to travel the length of the handle after the removal of the Bunsen burner. In case the handle does not continue to burn after the first ignition, the burner shall be placed under the free end for a period of 30 seconds immediately following the extinction of the flame. If the handle does not continue to burn after the second ignition and the flame becomes extinct before reaching the opposite end of the handle, the handle shall then be considered self-extinguishing. When tested as described, the plastic handle shall not burn at a rate exceeding 2 inches per minute.

4.6 **Rejected lots.**—Rejected lots may be offered again for inspection and tests provided the contractor has repaired or removed

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all nonconforming vises. Samples shall again be selected from such resubmitted lots and examined to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies should refer to Federal Standard No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation and packaging (unless otherwise specified (see 6.2)).

5.1.1 *Level A.*—Level A cleaning, preservation, and packaging shall be in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Level C cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing (unless otherwise specified (see 6.2)).

5.2.1 *Level A.*—Level A packing shall be in accordance with Military Specification MIL–P–15424.

5.2.2 *Level B.*—Level B packing shall be in accordance with Military Specification MIL–P–15424.

5.2.3 *Level C.*—Vises, preserved and packaged as specified in 5.1.1 and 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.*—Vises shall be marked for shipment in accordance with Military Specification MIL–P–15424.

5.3.2 *Nonmilitary agencies.*—Interior and exterior shipping containers shall be marked with the name, type, class, style, and size (as applicable) of the commodity, as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the manufacturer, and the number of the contract or order.

6. NOTES

6.1 Intended use.

6.1.1 Type I vises are intended for light bench work and are designed with an integral clamp for attaching to a bench or table.

6.1.2 Type II vises are intended for blacksmith work and are suitable for bench or table mounting.

6.1.3 Type III vises are intended for sheet-metal work and are suitable for bench or table mounting.

6.1.4 Type IV vises are intended for bench work and are suitable for bench or table mounting. Vises are designed with either integral or detachable pipe holding jaws.

6.1.5 Type V vises are intended for light work and are designed with a handle for holding in the hand.

6.1.6 Type VI vises are intended for light bench work and are suitable for bench or table mounting.

6.1.7 Type VII vises are intended for heavy bench work and are suitable for bench mounting.

6.1.8 Type IX vises are intended for holding small items which are usually round, and are suitable for holding in the hand. Vises are designed with a hollow handle on one end and a chuck on the other end.

6.1.9 Type X vises are intended for holding various sizes of pipe and are suitable for bench or pipe stand mounting.

6.1.10 Type XI vises are intended for holding various sizes of cable during splicing

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operations and are suitable for bench mounting.

6.1.11 Type XII vises are intended for holding handsaws during teeth sharpening operation and are suitable for bench or table mounting.

6.1.12 Type XIII vises are intended for holding band-saw blades during teeth sharpening operations and are suitable for bench or table mounting.

6.1.13 Type XIV vises are intended for holding circular saw blades during teeth sharpening operations and are suitable for bench or table mounting.

6.1.14 Type XV vises are intended for holding various sizes and shapes of wood and are suitable for bench or table mounting. Vises are designed so that the jaw top(s) is flush with the top of the bench or table when mounted.

6.1.15 Type XVI vises are intended for holding various sizes and shapes of wood and are suitable for bench or table mounting. Vises are designed to permit the jaws to be tilted from a vertical position through any angle up to 90 degrees and at the same time be capable of being rotated through 360 degrees.

6.1.16 Type XVII vises are intended for holding small work and are suitable for machine table mounting.

6.1.17 Type XVIII vises are intended for holding medium size work and are suitable for bench mounting. Vises are designed with a sub-base which, when removed, makes the vise suitable for machine table mounting.

6.1.18 Type XIX vises are intended to grip and hold square, round, tapered, or irregular shaped objects and are suitable for bench or machine table mounting.

6.2 **Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and style of vise required (see 1.2).

(*c*) Size, jaw width, or capacity required (see applicable table).

(*d*) Specify color required unless color is optional (see 3.2.2).

(*e*) Whether jaw caps are required (see 3.5.2 and 3.9.8.1); and if so, jaw widths required (see applicable tables) and material required (see 3.5.2).

(*f*) Whether pockets for pipe jaws are required (see 3.6).

(*g*) If the work contact area is not to be diamond or square serrated (see 3.8.3, 3.9.8, 3.21.2.3.6, and 3.23.3).

(*h*) Whether preservation and packaging should be by level A or C, and whether packing should be by level A, B or C (see section 5).

(*i*) Cleaning, preservatives, unit packaging, packing and marking, if different (see section 5).

6.2.1 When replacement parts are required, the procurement document should specify the following:

(*a*) The name or trade-mark of the manufacturer of the vise for which the part is required.

(*b*) Proper name or number of the part required.

(*c*) Quantity of each item required.

(*d*) Other data or description applicable to the vise which will assist in identifying the proper replacement parts.

6.3 **Superseding data.**—All the classes of Federal Specification GGG–V–436c have been changed from alphabetical to numerical sequence, and the styles from numerical to alphabetical sequence, in this specification. The nomenclature which was in conflict with the Federal Item Identification Guide for Supply Cataloging Handbook H6–1 has been changed as follows:

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GGG–V–436c	*This specification*
Type IV—Combination, bench and pipe, swivel base, stationary jaw.	Type IV—Bench and pipe (swivel base, stationary jaw).
Type VII—Machinists', bench.	Type VII—Machinist's.
Type XI—Riggers', splicing.	Type XI—Wire rope splicing.
Type XII—Saw.	Type XII—Handsaw filing.
Class A—Hand, with filing guide.	Class 1—With filing guide.
Class B—Hand, without filing guide.	Class 2—Without filing guide.
Class C—Band saw.	Type XIII—Band saw filing.
Class D—Circular saw.	Type XIV—Circular saw filing.
Type XIII—Woodworkers'.	Type XV—Woodworker's.
Class A—Continuous screw.	Class 1—Continuous screw.
Class B—Quick acting.	Class 2—Quick acting.
Class C—Universal.	Type XVI—Patternmaker's.
Type XIV—Precision.	Type XVII—Machine table.
Class A—Combination bench and machine.	Type XVIII—Bench and machine table.
Class B—Toolmakers'.	Type XVII, class 1—Toolmaker's.
Style 1—Slotted base.	Style A—Slotted base.
Style 2—Solid base.	Style B—Solid base.
Class C—Drill.	Type XVII, class 2—Drill.
Style 1—Quick screw.	Style A—Quick action.
Style 2—Continuous screw, light duty.	Style B—Continuous screw, light duty.
Style 3—Continuous screw, heavy duty.	Style C—Continuous screw, heavy duty.
Type XV—Contour gripping.	Type XIX—Contour gripping.

6.4 Transportation description.—Transportation description applicable to these items is:

Vises, iron or steel:
Carload weight 36,000 pounds.
Truckload volume weight 36,000 pounds.

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